

Generating University Timetable by using a Combination of Heuristic and Random Selection

Suryanti Awang¹

FSKKP

Universiti Malaysia Pahang

Pahang, Malaysia

suryanti@ump.edu.my

Mohamad Fairol², Marzuki Khalid³, Rubiyah Yusof⁴

AISB,

Universiti Teknologi Malaysia

Kuala Lumpur, Malaysia

mfz_peyo@yahoo.com^{2,3,4}

Abstract- The importance of timetable is to organize the arranged event in an organized list. The faculty or university timetable usually consists of huge data or event with a lot of constraint and characteristics that unable to change and need to be scheduled. The reliable and fast generation of timetable is demanding in generating the timetable especially if the timetable system is running in a web base. Unfortunately, there are a lot of timetable systems that unable to complete the generation in a short period of time. Here, we proposed a new combination of heuristic and random selection approach in generating the timetable with fast and reliable result. Our combination was experimented for one semester in Faculty of Electrical Engineering of Universiti Teknologi Malaysia that consists of huge number of students with parallel lectures, multi-disciplinary subjects, limited rooms and also lecturer's constraint. The performance reveals that this combination able to generate the timetable for the whole faculty within only 16 seconds, thus, verifying its effectiveness and reliability.

Keywords- *timetable, random selection, heuristic*

micro-GA has been applied to the timetable of the faculty [1]. The problem with the previous algorithm is we need a long period of time to generate the timetable.

In this paper, we focus on generating the timetable with heuristic and random selection. With the new algorithm, we hope to be able to generate the timetable within a short period of time. This is important of this research is to get variety pattern of solution for the timetable. Based on the previous experience by using micro-GA, the same pattern of generation might be happened. This situation will delay the timetable generation. Therefore with this research, the same pattern of timetable generation can be avoided. Other than that, we try to achieve a short period of time in generating the timetable since the timetable system is running in web-base. It is important for user satisfaction when they use this system without neglecting the reliability of the system.

Based on that, the objective of this research is to generate the timetable that consists of huge data within a short period of time generation. The data involved in the experiment are for one faculty with all level of student, from level 1 until level 5. The data is consist of the subjects offered for 1 (one) semester. A detail about the data is described in testing section.